

Model: Simple Linear Regression

The examples that i practiced are based on numpy and pandas
example number 1 and 2 are from numpy and third example is from pandas

Example 1: House Rent Prediction

****Predict monthly house rent based on house size in square feet.**

Data: House Size → Rent

```
House Size (sqft): [800, 900, 1000, 1200, 1500]
House Rent (₹):    [10000,12000,15000,18000,22000]
```

Predict rent for 1100 sqft. Solution: A straight line is fitted between house size and rent to predict new values.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Data
house_size = [800, 900, 1000, 1200, 1500]
rent = [10000, 12000, 15000, 18000, 22000]

# Convert to numpy
X = np.array(house_size).reshape(-1, 1)
y = np.array(rent)

# Model
model = LinearRegression()
model.fit(X, y)

# Prediction
new_size = 1100
predicted_rent = model.predict([[new_size]])

print("Predicted Rent:", predicted_rent[0])

# Plot
plt.scatter(house_size, rent, color='red')
plt.plot(house_size, model.predict(X), color='blue')
plt.scatter(new_size, predicted_rent, color='green')
```

```
plt.xlabel("House Size")
plt.ylabel("Rent")
plt.show()
```

Example 2: Salary Prediction

Predict employee salary based on years of experience.

Data: Experience → Salary

```
Experience (years): [1, 2, 3, 4, 5]
Salary (₹):          [20000, 30000, 40000, 50000, 60000]
```

Question: Predict salary for 3.5 years of experience. Solution: The regression line learned from past data is used to estimate salary.

```
# Step 1: Import libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Step 2: Dataset
experience = [1, 2, 3, 4, 5]
salary = [20000, 30000, 40000, 50000, 60000]

# Step 3: Convert to arrays
X = np.array(experience).reshape(-1, 1)
y = np.array(salary)

# Step 4: Train model
model = LinearRegression()
model.fit(X, y)

# Step 5: Prediction
new_exp = 3.5
predicted_salary = model.predict([[new_exp]])

print(f"Predicted salary for {new_exp} years experience: ₹
{predicted_salary[0]:.0f}")

# Step 6: Visualization
plt.figure(figsize=(8, 5))
plt.scatter(experience, salary, color="purple", s=100, label="Actual Data")
```

```
plt.plot(experience, model.predict(X), color="black", linewidth=2,
label="Regression Line")
plt.scatter(new_exp, predicted_salary, color="green", s=150, label="Predicted
Salary")
plt.xlabel("Years of Experience")
plt.ylabel("Salary (INR)")
plt.title("Salary Prediction using Linear Regression")
plt.legend()
plt.grid(True)
plt.show()
```

Example 3: YouTube Views Prediction

Predict total YouTube views based on first-hour likes.

Data: First Hour Likes → Total Views

```
Likes: [120, 340, 510, 230, 670]
Views: [8500, 22000, 41000, 15000, 53000]
```

Question: Predict views for 750 likes. Solution: Train a simple linear regression model and predict for new input.

```
# Step 1: Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Step 2: Create dataset
data = {
    'first_hour_likes': [120, 340, 510, 230, 670, 390, 150, 720],
    'total_views': [8500, 22000, 41000, 15000, 53000, 28000, 10500, 58000]
}

df = pd.DataFrame(data)

# Step 3: Prepare data
X = df[['first_hour_likes']]
y = df['total_views']

# Step 4: Train model
model = LinearRegression()
```

```
model.fit(X, y)

# Step 5: Prediction
new_likes = 750
predicted_views = model.predict([[new_likes]])

print(f"Predicted views for {new_likes} likes: {predicted_views[0]:.0f}")

# Step 6: Visualization
plt.figure(figsize=(8, 5))
plt.scatter(df['first_hour_likes'], df['total_views'],
            c=df['first_hour_likes'], cmap='viridis',
            s=100, label="Actual Data")

X_line = np.linspace(df['first_hour_likes'].min(),
                     df['first_hour_likes'].max(), 100).reshape(-1, 1)

plt.plot(X_line, model.predict(X_line),
         color="red", linewidth=2, label="Regression Line")

plt.scatter(new_likes, predicted_views,
            color="green", s=150, label="Predicted View")

plt.xlabel("First Hour Likes")
plt.ylabel("Total Views")
plt.title("YouTube Views Prediction")
plt.legend()
plt.grid(True)
plt.show()
```